

Example:

```

num = input("Enter a number")
if num > 0:
    print("Positive number")
elif num == 0:
    print("Zero")
else:
    print("negative number")

```

output

Enter a number: 7

Positive number

Enter a number: 0

zero

Enter a number: -1

Negative number

- Q. Write a python code to find the largest numbers among three integers.
- Q. Write a python code that the given no. is divisible by 5 or 10.
- Q. Write a python code to evaluate student mark system.
- Q. Write a python code to find the roots in the quadratic equation.
- Q. Write a python code to find traffic light system.
- Q. Write a python code

① Divisible by 5 or 10

```
n = int(input("Enter the number"))
if n % 10 == 0:
    print("The number is divisible by 10")
elif n % 5 == 0:
    print("The number is divisible by 5")
else:
    print("The number is not divisible by 5 or 10")
```

② Largest integer

```
a = int(input("Enter first number:"))
b = int(input("Enter second number:"))
c = int(input("Enter third number:"))
if a >= b and a >= c:
    print("Largest =", a)
elif b >= a and b >= c:
    print("Largest =", b)
else:
    print("Largest =", c)
```

③ Student mark system:

```
mark = int(input("Enter the mark:"))
if mark >= 90:
    print("Grade A")
elif mark >= 75:
    print("Grade B")
elif mark >= 50:
    print("Grade C")
elif mark >= 35:
    print("Pass")
else:
    print("Fail")
```

D	D	M	M	Y	Y	Y	Y

④ `a = int(input("enter a number:\n"))`
 if `a >= 0`:
`print("Positive no")`

⑤ `amount = int(input("enter purchase amount:\n"))`
 if `amount > 2000`
`amount = amount - 500`
`print("Final amount:" > amount)`

⑥ `category = input("enter a category:\n")`
 if `category == "Sports"`:
`print("bonus mark provided")`

⑦ `year = int(input("enter a year:\n"))`
 if `year % 4 == 0`
`print("leap year")`
 else:
`print("not leap year")`

⑧ `age = int(input("enter age:\n"))`
 if `age >= 18`
`print("eligible for voting")`
 else:
`print("not eligible for voting")`

9. `a = int(input("enter a no:\n"))`
 if `a % 2 == 0`
`print("even no")`
 else:
`print("odd no")`

10) `a = int(input("enter a no:"))`
`if a > 0:`
`print("Positive")`
`elif a == 0:`
`print("Zero")`
`else:`
`print("negative")`

16/7/26 Loop:

While loop:

initialization

while condition

// Body of the while

// true part

// increment / Decrement

Q. Display 1 to 10

Display 10 to 1

`i = 1`

`i = 10`

`while (i <= 10):`

`while i >= 1:`

`print(i)`

`print(i, end = " ")`

`i = i + 1`

`i = i - 1`

Q. Write a python code to print odd no. from 1 to 20

ans.

`i = 1`

`i = 1`

`while (i % 2 != 0):`

`while (i <= 20):`

`print(i)`

`if (i % 2 != 0)`

`i = i + 1`

`print(i)`

`i = i + 1`

Ans:

```
i = 1
while (i <= 20):
    if (i % 2) == 0:
        Print(i)
    i = i + 1
```

```
i = 1
while (i <= 20):
    if (i % 2) == 0:
        Print(i)
    i = i + 1
```

Q. Write a Python code to find the sum of series.

Ans:

```
n = int(input("enter n"))
i = 1
sum = 0
while (i <= n):
    sum = sum + i
    i = i + 1
Print("sum")
```

Q. Write a python code to find the factorial of the given number

```
n = int(input("enter n"))
i = 1
factorial = 1
while (i <= n):
    factorial = factorial * i
    i = i + 1
Print("factorial")
```

Q. Write a python code to find Fibonacci Series

```
0 1 1 2 3 5 8 13 21 34
i = 1
a = 1
b = 1
while i <
```

```

i = 0
a = 0
b = 1
while i <= 10:
    Print(a):
    c = a + b;
    a = b
    b = c
    i += 1

```

Q. Write a python code to find the sum of even fibonacci series

0, 2, 8, 34.

```

i = 1
a = 0
a = 0 Sum = 0
b = 1 Sum = 0
while i <= 10:
    if i % 2 == 0:
        Sum = Sum + i
        i += 1
    Print("Sum")

i = 1
a = 0
b = 1
Sum = 0
while (i <= 10):
    c = a + b
    a = b
    b = c
    if (a % 2 == 0):
        Sum = Sum + a
    i = i + 1

```

Q. Write a Python code to reverse the integer

```

c = 1
while (i <= 10)
    Print("Sum")

n = int(input("Enter a number"))
rev = 0
while (n > 0):
    a = n % 10
    rev = rev * 10 + a
    n = n // 10
Print("rev")

```

$n = 52$

while ($52 > 10$):

$a = 52 \% 10 = 2$

$rev = rev * 10 + a$

$n = 52 // 10$

while ($5 > 0$):

$a = 5 \% 10$

$rev = rev * 10 + a = 325$

$n = 5 // 10 = 0$

while ($10 > 0$): false

print ("rev") = 325

Q. Write a python code to check whether the given number is palindromic or not.

$n = 121$

$rev = 121$

$i = 1$

$i = n$

$rev = 0$

$org = n$

$sum = 0$

while ($n > 0$):

$a = n \% 10$

$rev = rev * 10 + a$

$n = n // 10$

if ($rev == org$):

print ("The given no is palin")

else

print ("The " " " " is not pal")

Q. find the sum of digits of a given number.

i = n

n = 523

rev = 0

while (n > 0):

a = n % 10 = 3

rev = rev * 10 + a = 3

n = n // 10 = 52

while (52 > 0):

a = 52 % 10 = 2

Q. Write a python code check whether the given no. is armstrong number or not.

i = n

n = 153

rev = 0

while (n > 0):

a = n % 10 = 3

rev = rev + a * * 3

n = n // 10 = 15

while (15 > 0):

a = n % 10 = 5

rev = rev + a * * 5

n = n // 10 = 1

while (1 > 0):

a = n % 10 = 1

rev = rev + a * * 1

n = n // 10 = 0

while (0 > 0): False

print("rev")

For Statement

```
for ch in python
    print("ch")
```

```
for num in [10, 20, 30]
    print("num")
```

```
For item in (1, 2, 3)
    print("item")
```

Range() function:

Syntax

range(n) - generate numbers from 0 to n-1;

eg: range(10) = 0 1 2 3 4 5 6 7 8 9

range(m, n) - generate numbers from m to n-1;

eg: range(4, 10) = 4, 5, 6, 7, 8, 9

eg: range(10, 10, 2)

Syntax

```
for i in range(end value):
```

```
// Body of the for
```

```
// true part.
```

```
for x in range(5):
```

```
    print(x) # 0 1 2
```

```
for x in range(1, 3):
```

```
    print(x) # 1, 2
```

```
for x in range(3, 0, -1):
```

```
    print(x) # 3 2 1
```

```
for x in range(5, 0, -2):
```

```
    print(x) # 5 3 1
```

```
for x in range(-4, 4):
```

```
    print(x) # -4 -3, -2 -1 0 1 2 3
```

```
For x in range(1,10,2):
    print(x) # 1 3 5 7 9
```

Functions

- function is a user defined piece of code
- it works only when it is called by user.

Built in function :-

len()

```
text = "python"
print(len(text))
```

max()

```
numbers = [10, 25, 8, 50, 15]
print(max(numbers))
```

Modules:

importing modules

→ import statement

→ from statement

```
from math import log
```

```
num = float(input("Enter a positive number"))
```

```
print("natural log" =, log(num))
```

Defined function

Function is a set of statement that perform specific function.

Void

```
c = a + b
```

```
print(c)
```

```
sum(10, 20) # function call
```

```
def Student(name, age=18):
    print("name", name)
    print("Age", age)
    Student("John")
    Student("Harry", 20)
```

output

Name: John
age: 18

Name: Harry
Age: 20

Python return more than one values.

```
# def sum_diff(x, y):
    add = x + y
    diff = x - y
    return add, diff
```

x = 9

y = 3

```
a, b = sum_diff(x, y)
```

```
print("sum = a")
```

```
print("diff = b")
```

```
# def greet_msg(name, msg):
    print(name, msg)
```

Calling function

```
def sum(*n)
```

```
    total = 0
```

```
    for i in n:
```

```
        total += i
```

```
    print("Sum = ", total)
```

calling function

Sum()

O/P Sum

Sum(10)

O/P Sum

Sum(10, 20, 30, 40)

O/P